Stratification Foundations: Identity and Utility (DRAFT)

Monique E. Davis, Ph.D.*

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Abstract

This note builds on identity-conditioned endowments (Note 1) to formalize individual decision-making under stratification. I develop a utility framework in which preferences incorporate not only private payoffs and risk, but also group identity welfare, group status, positional rank, and social penalties. Identity and power structures shape both the inputs and parameters of utility, generating systematic differences in behavior and outcomes across groups even when preferences are otherwise identical.

This note is part of the *Stratification Foundations* series (Notes 1–7), which develops a microeconomic framework of stratification from primitives to equilibrium and welfare.

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*Economist, W.E. Upjohn Institute for Employment Research. Email: davis@upjohn.org.

This note is part of an ongoing project to formalize a stratification economics framework to connect theory and empirics. Feedback welcome at davis@upjohn.org. Please do not cite without permission.

These ideas are inspired by, and build on, the robust existing stratification economics literature, to which the author is indebted. A full bibliography will be developed in future drafts.

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1 Group Identity and Individual Utility

Building on Note 1, where endowments are identity-conditioned, this note formalizes how individuals make decisions within stratified environments. I develop a utility framework in which identity and power structures shape both the inputs to decision-making (expected payoffs, risk, constraints) and the parameters governing preferences (group welfare, status, rank, and social penalties). This establishes how stratification affects not only opportunities, but also how individuals evaluate and respond to those opportunities.

Setup

Each individual i has:

- Private tastes θ_i,
- Group identity utility weight $\alpha_i \geq 0$ (weight on identity's welfare),
- Group identity status weight $\beta_i \geq 0$ (weight on identity's relative position in social hierarchy),
- Positional rank weights $\gamma_i \geq 0$ (weight on individual position amongst others),
- Institutional deviance weight $\delta_i \geq 0$ (weight on institutional sanctions),
- Identity norm deviation weight $\eta_i \geq 0$ (weight on the deviation from identity norms).

Self- and Perceived Identity

Each individual i has a **self-identity** $g_i \in G$, while others may hold beliefs $\hat{g}_i \in G$ about i's group identity. These beliefs need not coincide with i's self-identity.

Intuition: Identity matters not just in how individuals see themselves, but how others classify them. Differences between self- and perceived identity influence opportunities and constraints when individuals interact with other individuals or institutional agents.

Positional Rank Preferences:

Let Y denote the scalar outcome (e.g., income, wealth, education).

Define absolute and within-identity relative outcome percentiles as

$$l_i^{\text{abs}} \;=\; F_Y\left(y_i\right) \in [0,1]\,, \qquad l_i^{\text{within}} \;=\; F_{Y|g_i}\left(y_i \mid g_i\right) \in [0,1],$$

where $F(\cdot)$ represents percentile functions.

For cross-group rank, evaluate the individual against the reference identity's distribution using the *percentile gap*

$$l_i^{\text{cross}} \;:=\; F_{Y|g^\star}\left(y_i\right) \;-\; F_{Y|g_i}\left(y_i\right),$$

which compares where i would rank in some reference identity (e.g., a dominant group) $g^\star$, used for cross-group comparison, to where i ranks in their own identity g_i. In other words, where does an individual's outcome place them in the distribution of a reference identity.

Positional rank utility, or how much individuals care about "keeping up" with others, may be represented as

$$\Xi_i\,(l) \;=\; \gamma_i^{\text{abs}}\,\xi\Big(l_i^{\text{abs}}\Big) \;+\; \gamma_i^{\text{within}}\,\xi\Big(l_i^{\text{within}}\Big) \;+\; \gamma_i^{\text{cross}}\,\xi(l_i^{\text{cross}})\,,$$

where $\xi\,(\cdot)$ is increasing in l.[1] The weights $\gamma_i = \big(\gamma_i^{\text{abs}}, \gamma_i^{\text{within}}, \gamma_i^{\text{cross}}\big)$ may depend on group identity g and institutional context Υ, which determine whether absolute, within-identity, or cross-identity standing is most salient.[2]

Intuition: Individuals value, not only absolute outcomes, but also their standing amongst others. Absolute rank reflects position in the overall distribution, within-identity rank captures peer-relative standing, and the cross-identity percentile gap shows how one's rank compares against a reference identity. This captures how positional rank matters in addition to identity status.

Individual utility:

Information and expected outcomes. For a choice vector x_i, the unrealized payoff/outcome is drawn from an identity-conditioned distribution:

$$\tilde{y}_i(x_i; \Upsilon, \Pi_{g_i}),$$

where Π_{g_i} is the payoff schedule allocated to identity g_i by the power structure Υ. Payoffs are identity-conditioned: Π_{g_i} centers around average identity payoffs π_{g_i}.

Each individual forms expectations and perceived risk based on their information set $I_i = I(\iota_i)$, but these are draws from identity-conditioned pools (I_g):

$$\mu_i(x_i) \;=\; \mathbb{E}[\tilde{y}_i(x_i) \mid I_i, g_i]\,, \qquad \sigma_i^2(x_i) \;=\; \text{Var}[\tilde{y}_i(x_i) \mid I_i, g_i]\,.$$

Group identity status. Power structures Υ establish the hierarchy for social group G:

$$S(g; \Upsilon),$$

which captures the relative standing in the ordinal group identity ranking, or group identity status, for $g \in G$ under Υ. Higher S means greater social privilege or legitimacy for identity g_i.

Risk aversion and uncertainty. Individual risk aversion $r\,(\rho_i)$ is increasing in the amount of individual risk exposure in their environment, or the stakes tied to x_i, captured by ρ_i:

$$r(\rho_i)\,\sigma_i^2(x_i).$$

Risk aversion and uncertainty around expected outcomes interact to amplify the risk penalty associated with x_i. Individual risk exposure ρ_i is drawn from P_{g_i} centered around ρ_g, thus it is also identity-conditioned.

Institutional deviance penalty. Power structures Υ may classify x_i as "deviant" in a identity-dependent way:

$$d_i(x_i; g, \Upsilon) \;\geq 0,$$

[1] For tractability, I assume additive separability; extensions could allow interactions.

[2] γ is distinct from β in that the former represents the importance of individual rank and the latter, identity rank.

which captures detection/sanction/stigma asymmetries tied to g under Υ. Higher d means greater deviance associated with identity g_i

Identity norm deviation costs. Individuals may internalize identity norms, or face peer sanctions, that imposes psychic and social costs based on their group identity g_i:

$$\psi_i(\|x_i - \bar{x}\|; g_i),$$

which captures penalties from deviating from the identity-norm vector $\bar{x}$ associated with identity g_i. The greater the deviation from norms, the higher the psychic/social costs.

Individual utility maximization problem. Given group identity g_i, endowments $\omega_i = (\kappa_i, \iota_i, \lambda_i, \rho_i, \pi_i)$, and preferences θ_i, the individual chooses an x_i^* that[3]

$$\begin{aligned}
\max_{x_i^* \in K(\kappa_i)} U_i(x_i) \;=\; & \underbrace{u(\mu_i(x_i))}_{\text{expected private payoff}} + \underbrace{\alpha_i\, U_{g_i}}_{\text{group identity welfare}} + \underbrace{\beta_i\, S(g_i)}_{\text{group identity status}} + \underbrace{\Xi_i(l_i\,(\mu_i(x_i)))}_{\text{positional rank}} \\
& - \underbrace{r(\rho_i)\,\sigma_i^2(x_i)}_{\text{risk penalty}} - \underbrace{\delta_i\, d_i(x_i)}_{\text{deviance penalty}} - \underbrace{\eta_i\, \psi_i(\|x_i - \bar{x}\|)}_{\text{norm deviation costs}} \\
& \text{s.t.} \qquad a(\lambda_i) \cdot x_i \;\leq\; b(\lambda_i).
\end{aligned}$$

where:

- $\omega_i \sim D_g\,(\mu = \Omega_g, \Sigma_g)$.
- $u\,(\cdot)$ is a concave utility function.
- Utility weights $(\alpha_i, \beta_i, \gamma_i, \delta_i, \eta_i)$ capture the relative importance of group identity, status, rank, and social penalties in individual decision-making.
- $U_g = \frac{1}{|g|} \sum_{j:g_j=g} u\,(\cdot\,; \theta_j)$ or another identity-utility aggregator depending on payoffs, group identity factors, or penalties.
- $a(\lambda_i)$ is the price schedule/cost structure for x_i shaped by $\Lambda_i = \Lambda(\lambda_i)$
- $b(\lambda_i)$ is individual income/opportunity ceiling shaped by Λ_i

Rearranging terms to highlight the components of individual utility under stratification:

$$\begin{aligned}
U_i(x_i) \;=\; & \underbrace{u(\mu_i(x_i)) \;-\; r(\rho_i)\,\sigma_i^2(x_i)}_{\text{private payoffs and risks}} \\
& + \underbrace{\alpha_i\, U_{g_i} \;+\; \beta_i\, S(g_i)}_{\text{group identity}} \\
& + \underbrace{\Xi_i(l_i(\mu_i(x_i)))}_{\text{positional rank}} \\
& - \underbrace{\delta_i\, d_i(x_i) \;-\; \eta_i\, \psi_i(\|x_i - \bar{x}\|)}_{\text{social penalties}}
\end{aligned}$$

[3]For readability, I suppress dependence on g_i and Υ in the utility function.

Benchmark UMP (individualist view). Taking the individualist view, if we assume every individual draws from identical economic endowment pools and that group identity is irrelevant so group identity factors, individual rank, and deviation penalties do not exist, we are left with the benchmark utility maximization problem:

$$\max_{x_i^{*\,\text{pop}} \in K(\kappa_i^{\text{pop}})} U_i(x_i^{\text{pop}}) \;=\; \underbrace{u(\mu_i(x_i^{\text{pop}}))}_{\text{expected private payoff}} \;-\; \underbrace{r(\rho_i^{\text{pop}})\,\sigma_i^2(x_i^{\text{pop}})}_{\text{risk penalty}}$$

$$\text{s.t.} \qquad a(\lambda_i^{\text{pop}}) \cdot x_i^{\text{pop}} \;\leq\; b(\lambda_i^{\text{pop}}).$$

Intuition: Individuals care about the private benefits and costs of their choices. They also care about group identity welfare, group identity status in the hierarchy, their absolute and relative rank, and the psychic costs of identity norm deviation. Stratification, by identity-conditioning the given endowments with more or less favorable factors, influences the salience of these factors and how the decisions individuals make will benefit or harm them. Thus, average outcomes for members of dominant identities are set up to consistently outperform the average outcomes for members of subaltern identities.

Compact Intuition: Group Identity and Individual Utility

Setup/Inputs: Endowments are identity-conditioned. Power structures define group hierarchies (S), shape payoff distributions (Π_g), and attach sanctions to identity-dependent deviance. Individuals possess group identities, care about their absolute and relative ranks, and worry about peer sanctions from norm deviation. Importantly, subaltern identities draw systematically tighter constraints (λ_i), noisier information (ι_i), and lower payoff multipliers (π_i), so even identical choices and ability can produce unequal returns.

Choices/Interactions: Individuals weigh (i) private outcomes and risk, (ii) group identity welfare and group identity status, (iii) positional rank (absolute, within-identity, cross-identity), and (iv) penalties from institutions and peers, subject to opportunity frontiers shaped by identity-conditioned feasibility, information, and payoffs.

Outcomes/Solution: Inequality persists not only because endowments are stratified, but because of the downstream impact on the parameters and inputs to utility. Thus, individuals with identical private preferences and abilities, but different group identities, systematically face unequal opportunity sets, expectations, and outcomes.

Table 1: Comparing Perspectives on Individual Utility: Individualist vs. Structuralist

Dimension	**Individualist Perspective**	**Structuralist Perspective**
Power Structures	Institutions are identity-neutral arbiters, or irrelevant.	Υ defines hierarchies S, stratifies endowments, and imposes sanctions.
Group Identities	Irrelevant to utility.	Self-identity g_i and perceived identity $\hat{g}_i$ alter opportunities, expectations, and penalties.
Endowments/Inputs	Single endowment pool.	Endowments are identity-conditioned: choice sets, information, constraints, risk, and payoffs vary by identity.
Decision Rules	Maximize private payoffs net of risk.	Utility includes private payoffs, group identity welfare/status, positional rank, and social penalties.
Outcomes	Identical choices yield identical payoffs, conditional on ability. Remaining inequality is typically interpreted as arising from differences in ability, effort, or culture.	Identical choices, conditional on ability, yield unequal payoffs if group identity endowments or hierarchies differ.

Bringing It Together

The structuralist framework generalizes the individualist UMP. Instead of assuming that identity is irrelevant and individuals maximize only private payoffs subject to prices and uncertainty, we recognize that identity and power structures shape the inputs and parameters of individual utility. E.g., two entrepreneurs with identical ideas and ability, but one faces systematically lower credit ceilings and higher perceived deviance risk, which alters their decision making. Individuals weigh private outcomes alongside group identity welfare, group identity status, rank, and social penalties. E.g., someone may vote against a local tax levy if it undermines their group identity's status, even if it would benefit their children's education. Because endowments are identity-conditioned, individuals with identical preferences θ_i and abilities, but different group identities, face systematically unequal outcomes.